

Sharp Initial Thresholds for Abelian-Bordered Binary Infinite Words

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Abstract

Let $\mu_{\text{ab}}(x)$ denote the maximum length of an Abelian-unbordered factor of a binary infinite word x , with value ∞ when these lengths are unbounded. Charlier, Harju, Puzynina and Zamboni asked whether $\mu_{\text{ab}}(x) < \infty$ forces x to be Abelian ultimately periodic. We determine the first sharp threshold layers. If $\mu_{\text{ab}}(x) \leq 13$, then x is ultimately periodic in the ordinary sense and its eventual period can be chosen of length at most 13. Both the threshold and the period bound are sharp: the least finite value of μ_{ab} attained by a non-ultimately-periodic binary word is 14. If $\mu_{\text{ab}}(x) \leq 14$, then x is Abelian ultimately periodic with Abelian period length at most 14, again sharply.

The upper bounds are computer-assisted but exact. We classify all recurrent cores of two finite overlap graphs. At threshold 14, every cyclic strongly connected component is a directed cycle of length at most 13. At threshold 15, all but four cyclic components are directed cycles, and the four branching components are absorbed by closed Parikh phases of lengths 4 or 14. A separate exact computation establishes the sharp period bounds.

We also prove a non-computational sufficient condition valid at arbitrary thresholds. If the frequency of 1 in x is the reduced fraction p/q , $M = \mu_{\text{ab}}(x) < \infty$, and $M < 2q$, then x is Abelian ultimately periodic with blocks of length q and weight p . The general question remains open outside these regimes.

Keywords. Abelian border; infinite word; ultimate periodicity; combinatorics on words; finite overlap graph; computer-assisted proof.

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1 Introduction

A nonempty finite word is bordered when it has a nonempty proper prefix equal to a suffix. Ehrenfeucht and Silberger [3] proved that an infinite word is purely periodic exactly when it has only finitely many unbordered factors. Replacing equality by Abelian equivalence gives a natural commutative analogue: two words are Abelian equivalent when they have the same Parikh vector, and a word is Abelian bordered when a nonempty proper prefix is Abelian equivalent to a suffix.

Charlier, Harju, Puzynina and Zamboni [1] asked whether a binary infinite word having only finitely many Abelian-unbordered factors must be Abelian ultimately periodic. They constructed aperiodic words all of whose factors of length at least 15 are Abelian bordered, but their examples remain Abelian periodic. The problem was still listed as open in the survey of Fici and Puzynina [2, Problem 42].

For a binary infinite word x , define

$$\mu_{\text{ab}}(x) = \sup\{|u| : u \text{ is an Abelian-unbordered factor of } x\}.$$

Thus the open problem asks whether $\mu_{\text{ab}}(x) < \infty$ implies Abelian ultimate periodicity. Our first result determines the exact initial boundary at which ordinary ultimate periodicity can fail.

Theorem 1.1. *If $\mu_{\text{ab}}(x) \leq 13$, then x is ultimately periodic and its eventual period can be chosen of length at most 13. Both constants are sharp. In particular,*

$$\min\{\mu_{\text{ab}}(x) : x \text{ is not ultimately periodic and } \mu_{\text{ab}}(x) < \infty\} = 14.$$

The first layer permitting ordinary aperiodicity still satisfies the desired Abelian conclusion.

Theorem 1.2. *If $\mu_{\text{ab}}(x) \leq 14$, then x is Abelian ultimately periodic with Abelian period length at most 14. The period bound 14 is sharp.*

Theorems 1.1 and 1.2 are proved by finite certificates which cover every infinite word satisfying the corresponding threshold condition; they are not bounded-prefix experiments. Our second type of result comes from the rational frequency and bounded discrepancy forced by the hypothesis.

Theorem 1.3. *Let $M = \mu_{\text{ab}}(x) < \infty$, and write the frequency of 1 as the reduced fraction p/q . If $M < 2q$, then x is Abelian ultimately periodic. More precisely, after deleting a finite prefix, x factors into consecutive blocks of length q , each containing exactly p occurrences of 1.*

Theorem 1.3 covers an infinite parameter regime but not the whole open problem. The remaining difficulty begins when the extremal discrepancy layer can have several possible return lengths.

2 Preliminaries

Let $x = x_0x_1x_2 \cdots \in \{0, 1\}^{\mathbb{N}}$. For a finite binary word u , its Parikh vector is

$$\mathcal{P}(u) = (|u|_0, |u|_1).$$

Words u, v are Abelian equivalent, written $u \sim_{\text{ab}} v$, when $\mathcal{P}(u) = \mathcal{P}(v)$. A word u of length L has an Abelian border of length k if $1 \leq k < L$ and its prefix and suffix of length k are Abelian equivalent. If $k > L/2$, deleting the common overlap from the two ends produces an Abelian border of length $L - k < L/2$. It is therefore enough to consider

$$1 \leq k \leq \lfloor L/2 \rfloor.$$

An infinite word is ultimately periodic if it has the form uv^ω . It is Abelian ultimately periodic if it admits a factorization

$$x = uv_0v_1v_2 \cdots, \quad v_i \sim_{\text{ab}} v_j \quad (i, j \geq 0).$$

The common blocks consequently have the same length and the same number of ones. We call that common length an Abelian period length.

For integers $C \leq N$, call a binary word of length N (C, N) -admissible if every one of its factors whose length lies in $[C, N]$ is Abelian bordered.

3 Finite overlap certificates

3.1 The overlap graph

Define a finite directed labeled graph $G_{C,N}$ as follows. Its vertices are all length- $(N-1)$ prefixes and suffixes of (C, N) -admissible words. Every admissible word $w = w_0w_1 \cdots w_{N-1}$ gives the edge

$$w_0 \cdots w_{N-2} \longrightarrow w_1 \cdots w_{N-1},$$

labeled by w_{N-1} .

If every factor of an infinite word x of length at least C is Abelian bordered, then its consecutive length- N windows form an infinite path in $G_{C,N}$. Since the condensation of a finite directed graph is acyclic, the path eventually remains inside a strongly connected component containing a directed cycle.

Lemma 3.1. *Let H be a finite strongly connected directed simple graph. If the number of internal edges of H equals the number of vertices, then H is a directed cycle.*

Proof. Every vertex of a strongly connected graph has positive internal indegree and outdegree. The sums of the internal indegrees and outdegrees both equal the number of edges. Equality with the number of vertices forces every internal indegree and outdegree to be one. Thus H is a directed cycle. $\square$

An infinite path which eventually remains in such a component has an ultimately periodic label sequence.

3.2 Closed Parikh phases

Branching components require a weaker certificate. Let H be a strongly connected component, let $\ell \geq 1$, and let $0 \leq m \leq \ell$. A set $Q_{\ell,m} \subseteq V(H)$ is a closed (ℓ, m) -phase if every length- ℓ path in H beginning in $Q_{\ell,m}$ has label weight m and terminates in $Q_{\ell,m}$.

Lemma 3.2. *Every infinite path in H which visits $Q_{\ell,m}$ factors from that visit onward into label blocks of length ℓ and weight m .*

Proof. Apply the two defining properties successively to consecutive length- ℓ path segments. Each segment has the same binary Parikh vector $(\ell - m, m)$, and its terminal state is again in the phase. $\square$

For fixed (ℓ, m) , the verification program computes the greatest closed phase by iterating the monotone operator

$$\Phi_{\ell,m}(Q) = \{v \in V(H) : \text{every length-}\ell \text{ path from } v \text{ has weight } m \text{ and ends in } Q\}$$

downward from $V(H)$. Stabilization is exact because H is finite. Several closed phases may be used at once.

Lemma 3.3. *Suppose Q is a union of closed Parikh phases in H , and the subgraph induced by $V(H) \setminus Q$ is acyclic. Then every infinite path in H is Abelian ultimately periodic.*

Proof. An infinite path cannot remain forever in a finite acyclic graph, so it eventually enters one of the closed phases contained in Q . Lemma 3.2 then gives an Abelian periodic factorization of its label tail. $\square$

These lemmas reduce both upper bounds to exact finite graph statements.

4 Exact classification of the first two layers

The admissible words and overlap graphs were constructed independently in C++ and Python. The Python verification also exhaustively compares the two definitions of Abelian borderedness on all 131,070 nonempty binary words of length at most 16, and checks small overlap graphs by brute force.

Proposition 4.1. *For $(C, N) = (14, 36)$, the graph $G_{14,36}$ has*

<i>admissible length-36 words</i>	38,450,
<i>vertices</i>	38,544,
<i>cyclic strongly connected components</i>	1,377.

Every cyclic strongly connected component is a directed cycle, and the largest such cycle has 13 states.

Proof of the upper bound in Theorem 1.1. If $\mu_{\text{ab}}(x) \leq 13$, every factor of length at least 14 is Abelian bordered. The length-36 windows of x therefore form an infinite path in $G_{14,36}$. Eventually the path remains in a cyclic strongly connected component. Proposition 4.1 and Lemma 3.1 show that this component is a directed cycle of length at most 13. Hence x is ultimately periodic with eventual period length at most 13. $\square$

Proposition 4.2. *For $(C, N) = (15, 30)$, the graph $G_{15,30}$ has*

<i>admissible length-30 words</i>	199,420,
<i>vertices</i>	217,280,
<i>cyclic strongly connected components</i>	2,540.

Exactly four cyclic components branch. Two have 40 vertices and 42 edges; in each, closed phases include a phase of length 4 and weight 2, their union covers 29 vertices, and the uncovered induced subgraph is acyclic. The other two have 60 vertices and 62 edges; closed phases include a phase of length 14 and weight 7, their union covers 54 vertices, and the uncovered induced subgraph is acyclic.

The other 2,536 cyclic components are directed cycles. A separate exact cycle-label post-processing computation shows that every one of these cycles has an Abelian period length at most 14.

Proof of the upper bound in Theorem 1.2. If $\mu_{\text{ab}}(x) \leq 14$, the length-30 windows of x form an infinite path in $G_{15,30}$. If its eventual component is one of the four branching components, Proposition 4.2 and Lemma 3.3 give a block factorization of length 4 or 14. Otherwise the eventual component is a directed cycle whose label word has an Abelian period of length at most 14. In all cases x is Abelian ultimately periodic with an Abelian period length at most 14. $\square$

The numerical assertions in Propositions 4.1 and 4.2 are certificate-bearing finite claims. The verification entry points and fixed outputs are described in Section 7.

5 Sharpness

5.1 The threshold for ordinary periodicity

Charlier et al. [1, Proposition 2 and Remark 2] use the two codewords

$$a = 010100110011, \quad b = 0101001100110011.$$

Any infinite concatenation of a and b has every factor of length at least 15 Abelian bordered. Choosing a non-ultimately-periodic codeword selector produces a word which is not ultimately periodic: in the aligned length-4 block alphabet, the block 0101 marks the beginning of a codeword and is followed by two or three copies of 0011, so the selector can be recovered from the binary word.

The same construction contains the factor

$$u = 01001100110011$$

of length 14. For border lengths $k = 1, \dots, 7$, the numbers of ones in the corresponding prefixes and suffixes are respectively

$$(0, 1), (1, 2), (1, 2), (1, 2), (2, 3), (3, 4), (3, 4).$$

Thus no such k is an Abelian border. A border longer than 7 would, after deleting the common overlap, give a shorter Abelian border. Hence u is Abelian unbordered and the constructed word has $\mu_{\text{ab}} = 14$. Together with Theorem 1.1 this proves the exact threshold assertion.

5.2 Sharp period bounds

Let $w = 10^{12}$. The periodic word w^ω has least ordinary period 13: for every $1 \leq p < 13$, its symbol at position 0 differs from the symbol at position p . Every factor of length $L \geq 14$ has the ordinary border of length $L - 13$. On the other hand, w is Abelian unbordered because for $1 \leq k \leq 6$ its prefix of length k has one occurrence of 1, while its suffix has none. Therefore

$$\mu_{\text{ab}}(w^\omega) = 13,$$

and the period-length bound in Theorem 1.1 is sharp.

Similarly, put $v = 10^{13}$. Then v is Abelian unbordered, while every factor of v^ω of length at least 15 is ordinarily bordered. Hence

$$\mu_{\text{ab}}(v^\omega) = 14.$$

Every block in an Abelian periodic factorization of a tail of v^ω must have the word's frequency $1/14$, so its length is a multiple of 14. Thus the least Abelian period length is 14, proving sharpness in Theorem 1.2.

6 A frequency-denominator criterion

We now prove Theorem 1.3. Let

$$A(n) = \sum_{j=0}^{n-1} x_j.$$

Having only finitely many Abelian-unbordered factors implies having only finitely many weakly Abelian-unbordered factors, as used in the proof of [1, Theorem 5]. Lemmas 3 and 4 of [1] therefore imply that x is balanced and has rational letter frequencies. Write the frequency of 1 in lowest terms as p/q and define the integer discrepancy

$$z_n = qA(n) - pn.$$

The bounded-deviation formulation of balance in [1, Proposition 1] implies that the set $\{z_n : n \geq 0\}$ is finite.

Lemma 6.1. For $i \geq 0$, $L \geq 1$, and $1 \leq k \leq L/2$, the factor $x[i, i+L)$ has an Abelian border of length k if and only if

$$z_{i+k} + z_{i+L-k} = z_i + z_{i+L}. \quad (6.1)$$

Proof. The prefix and suffix of length k contain the same number of ones exactly when

$$A(i+k) - A(i) = A(i+L) - A(i+L-k).$$

Substituting $qA(n) = z_n + pn$ cancels all linear terms and gives (6.1). $\square$

Put $M = \mu_{\text{ab}}(x)$ and $C = M + 1$. Every factor of length at least C is Abelian bordered. Choose a value R which occurs infinitely often among the z_n and is maximal among the infinitely recurrent values. After deleting a finite prefix, no value larger than R occurs. Write the successive occurrences of R as

$$s_0 < s_1 < s_2 < \dots$$

and set $g_i = s_{i+1} - s_i$.

Lemma 6.2. For every i ,

$$1 \leq g_i < C \quad \text{and} \quad q \mid g_i.$$

Proof. Suppose $g_i \geq C$. Apply the Abelian-border condition and Lemma 6.1 to the factor $x[s_i, s_{i+1})$. For some $1 \leq k \leq g_i/2$,

$$z_{s_i+k} + z_{s_{i+1}-k} = z_{s_i} + z_{s_{i+1}} = 2R.$$

Both terms on the left are at most R , so both equal R . At least one of the indicated positions lies strictly between s_i and s_{i+1} , contradicting their consecutiveness. Thus $g_i < C$.

Equal discrepancy at the endpoints gives

$$0 = z_{s_{i+1}} - z_{s_i} = q(A(s_{i+1}) - A(s_i)) - pg_i.$$

Since $\gcd(p, q) = 1$, it follows that $q \mid g_i$. $\square$

Proof of Theorem 1.3. Assume $M < 2q$. Then $C = M + 1 \leq 2q$. Write $g_i = r_i q$, where r_i is a positive integer. Lemma 6.2 gives

$$0 < r_i = \frac{g_i}{q} < \frac{C}{q} \leq 2,$$

and hence $r_i = 1$. Thus every $g_i = q$.

For every i , the block $x[s_i, s_{i+1})$ has length q , and equality of endpoint discrepancies yields

$$q |x[s_i, s_{i+1})|_1 - pq = 0.$$

Every such block therefore contains exactly p ones. These consecutive blocks cover the tail beginning at s_0 , and they all have Parikh vector $(q-p, p)$. This is an Abelian ultimately periodic factorization with period length q . $\square$

When $M \geq 2q$, the maximal discrepancy layer may have several return lengths $q, 2q, \dots$. The argument above no longer selects a single block length. Controlling only that extremal layer is not sufficient in general; the unresolved problem requires compatibility with the other discrepancy levels.

7 Verification and reproducibility

The finite results can be reconstructed from the verification material distributed with this manuscript. The principal checks are:

1. exhaustive agreement of direct and prefix-sum Abelian-border tests on all nonempty binary words of length at most 16;
2. brute-force controls for $(C, N) = (3, 6), (4, 8), (5, 10)$;
3. independent C++ and Python reconstruction of the $G_{14,36}$ and $G_{15,30}$ component certificates;
4. greatest-fixed-point computation of the closed Parikh phases in the four branching components and acyclicity of the uncovered subgraphs;
5. exact cyclic ordinary-period and Abelian-period computations, including the two sharp witnesses.

From the project entry directory, run

```
python -X utf8 verification\verify.py
python -X utf8 verification\verify_period_bounds.py
```

The C++17 sources are `verification\verify_threshold14.cpp` and `verification\verify_threshold15.cpp`. The programs use exhaustive finite graph traversal; no random sampling, floating-point comparison, SAT solver, or unverified external dataset enters the stated certificates.

8 Scope and prior-work boundary

The present results do not solve the full question of Charlier et al. They establish the sharp initial ordinary-periodicity threshold, the next Abelian periodicity layer with a sharp period bound, and a sufficient condition depending on the denominator of the rational frequency.

A targeted search of the original paper, the 2022 survey, and indexed follow-up literature found no result directly covering these statements. This is an internal novelty assessment, not a claim of exhaustive global priority. The manuscript has not undergone external expert review or formal peer review.

Filimonova and Puzynina [4] recently characterized Abelian periodicity for purely morphic binary words. Their object class and hypotheses differ from the Abelian-border threshold conditions considered here.

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